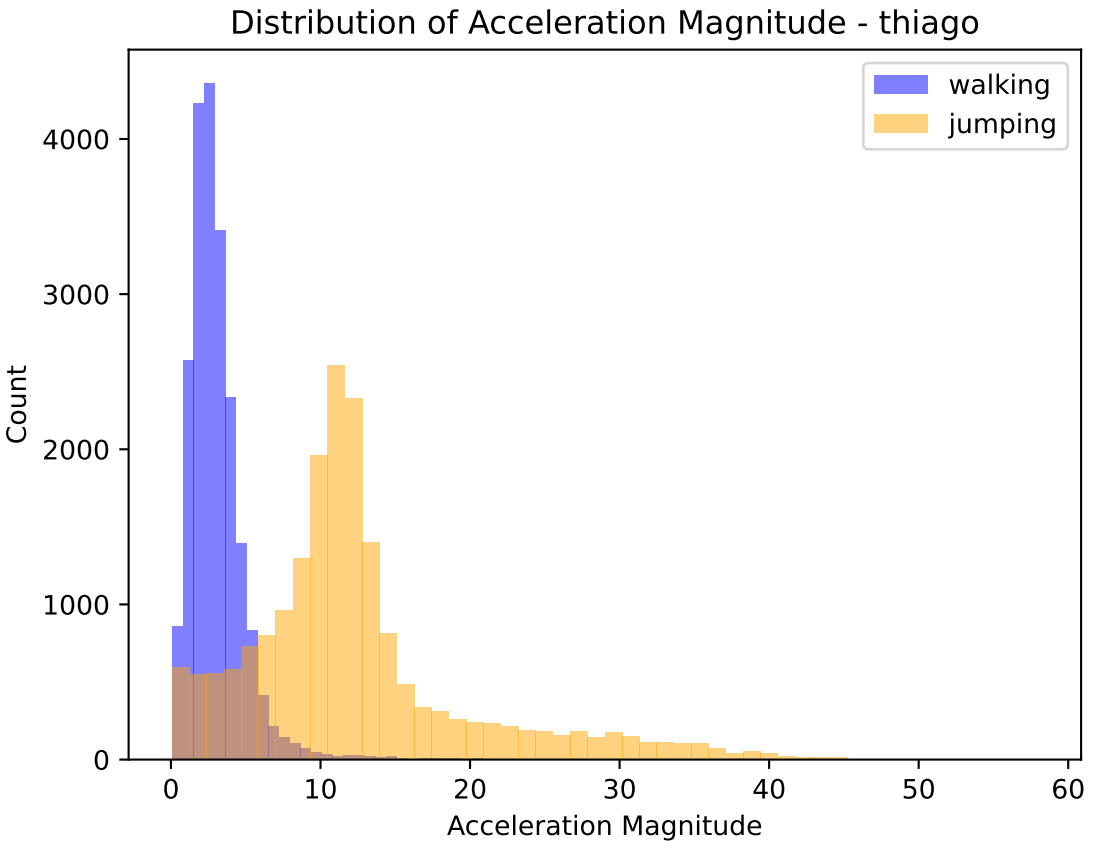
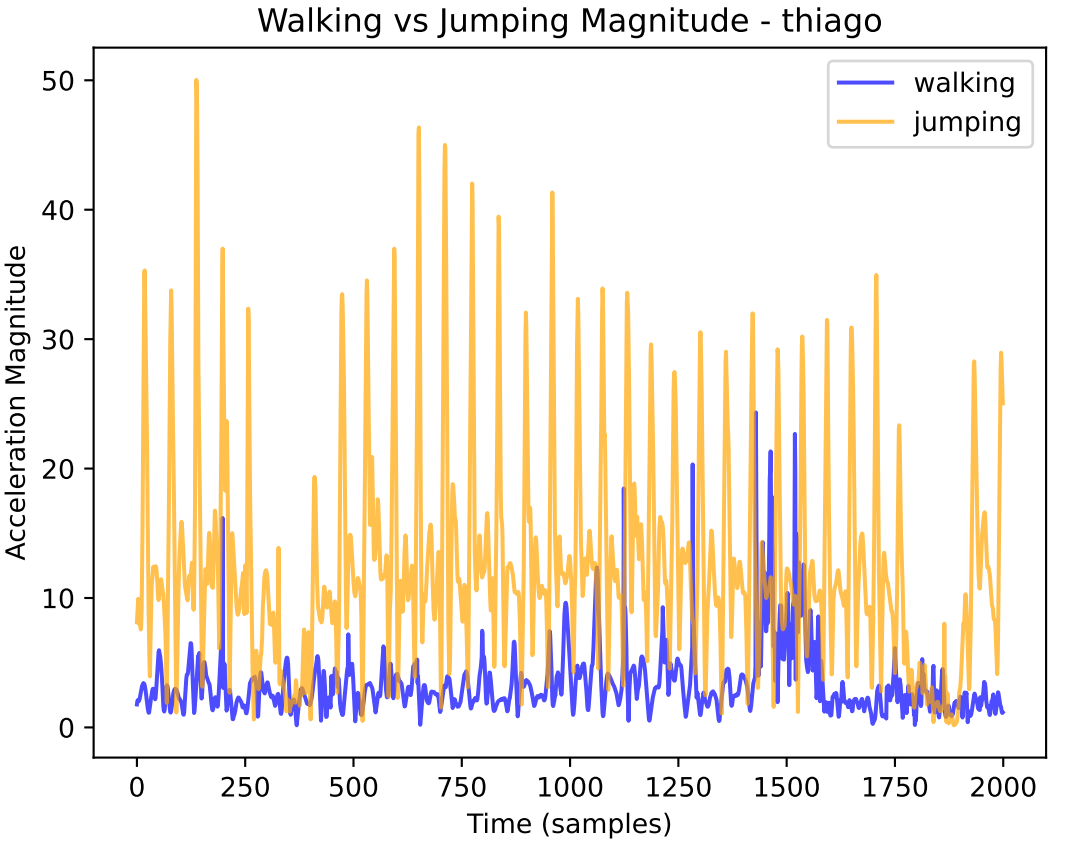
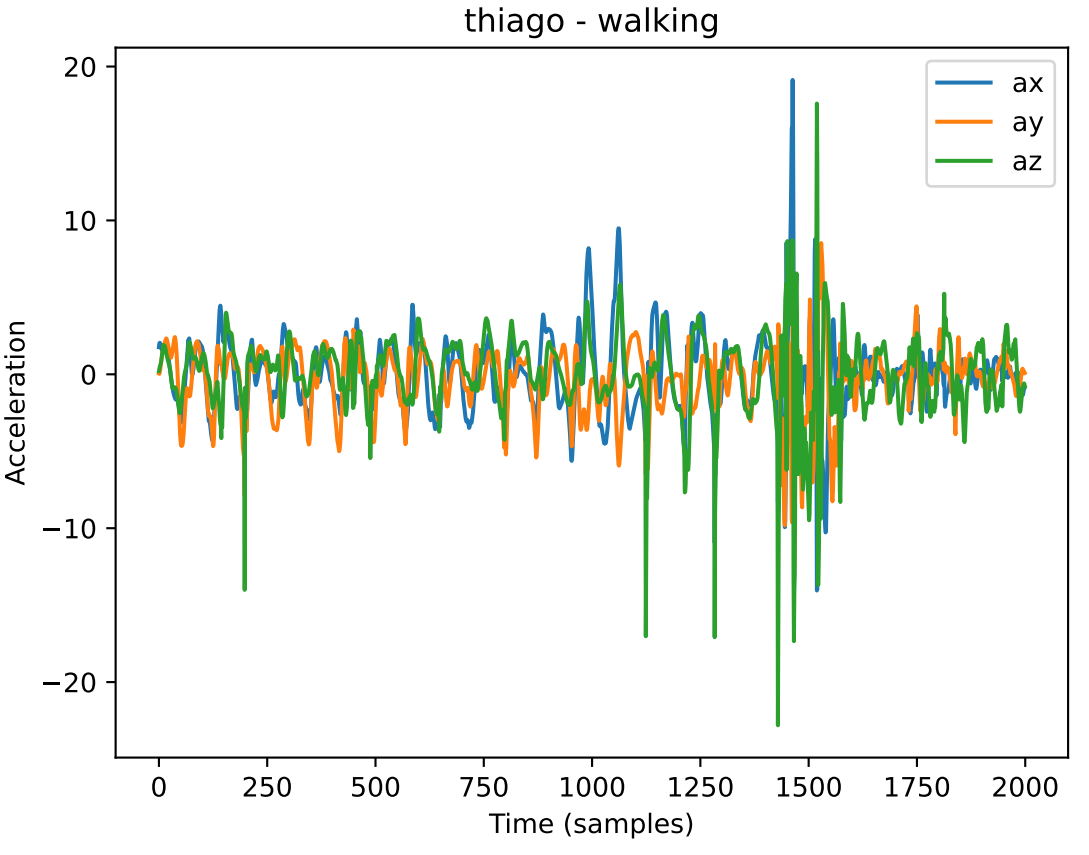
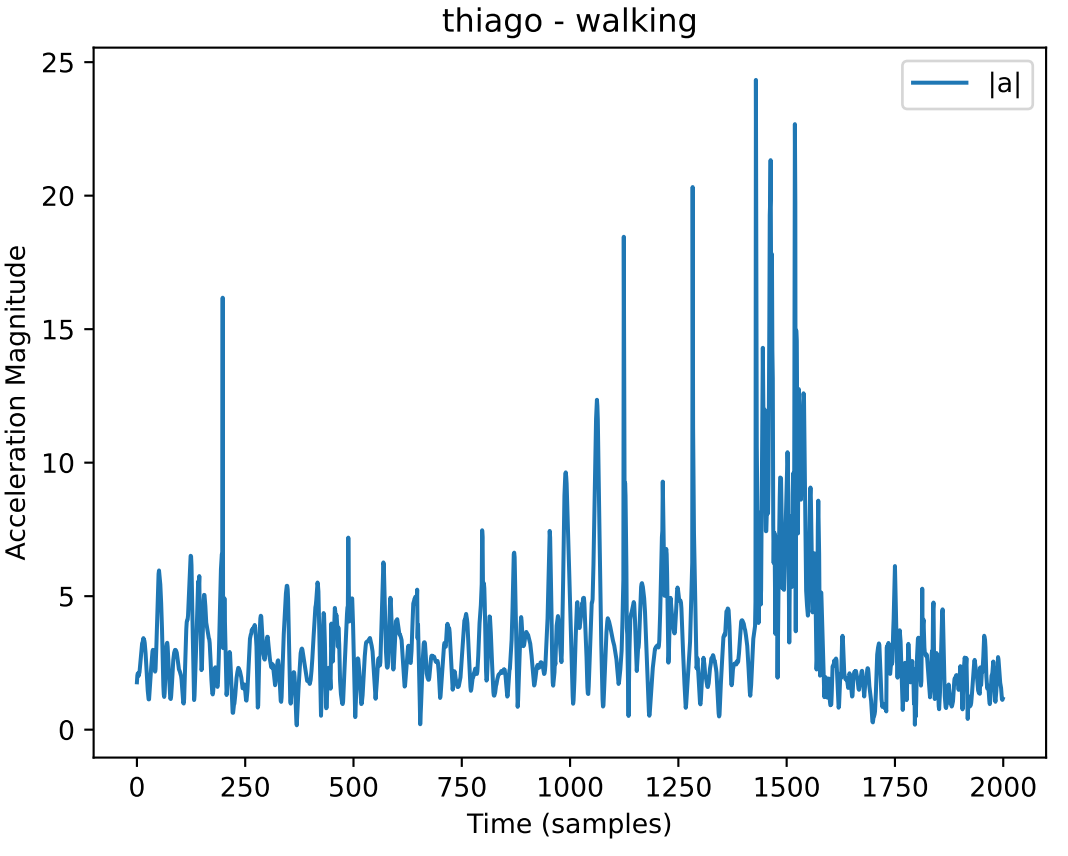
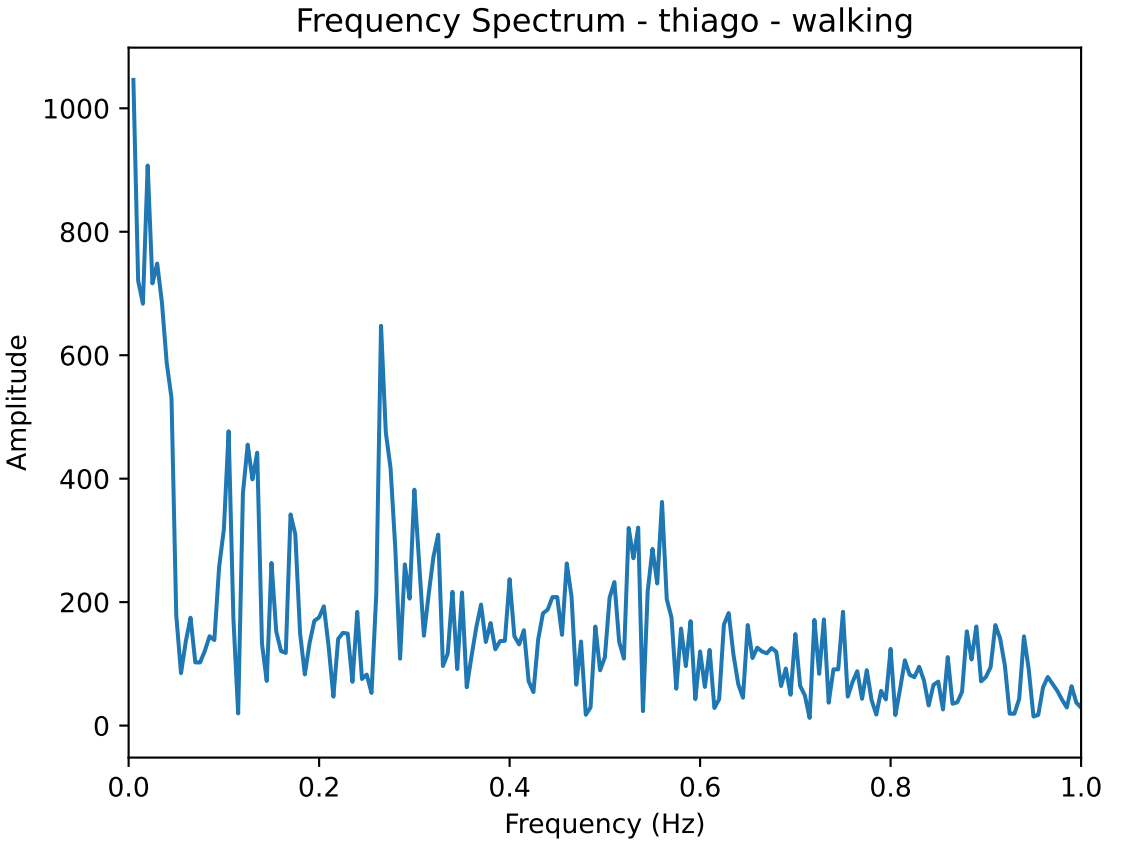
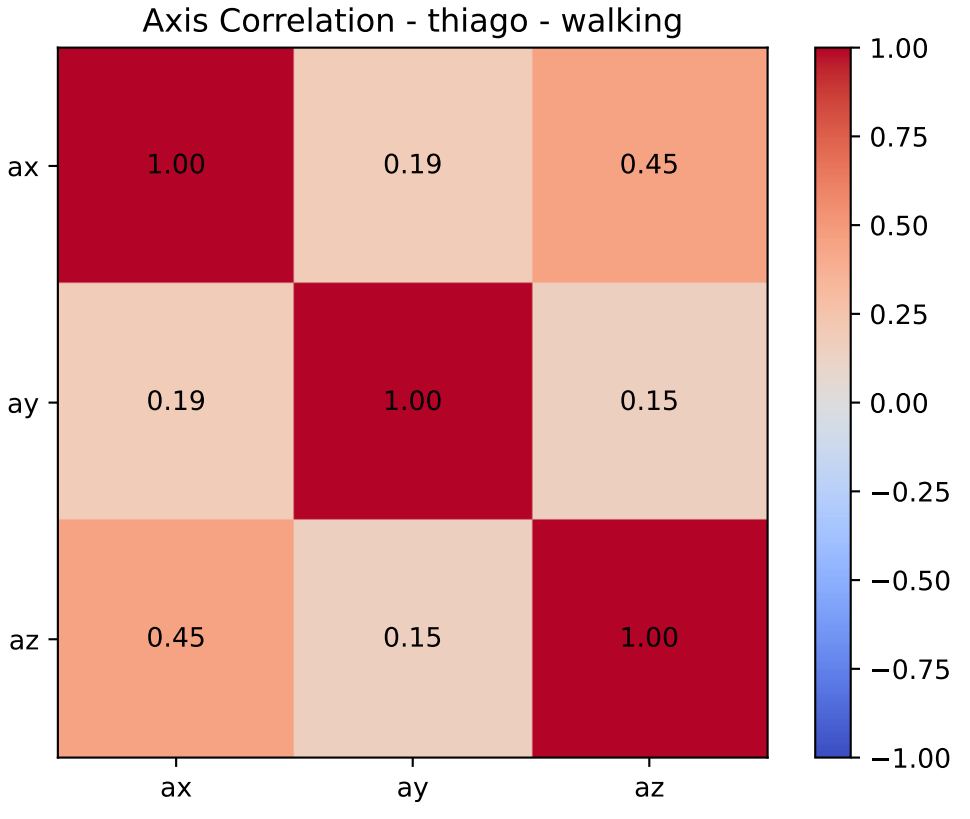

Distribution of Acceleration Magnitude - thiago
walking
jumping
Count
4000
3000
2000
1000
0
0
10
20
30
40
50
60
Acceleration Magnitude

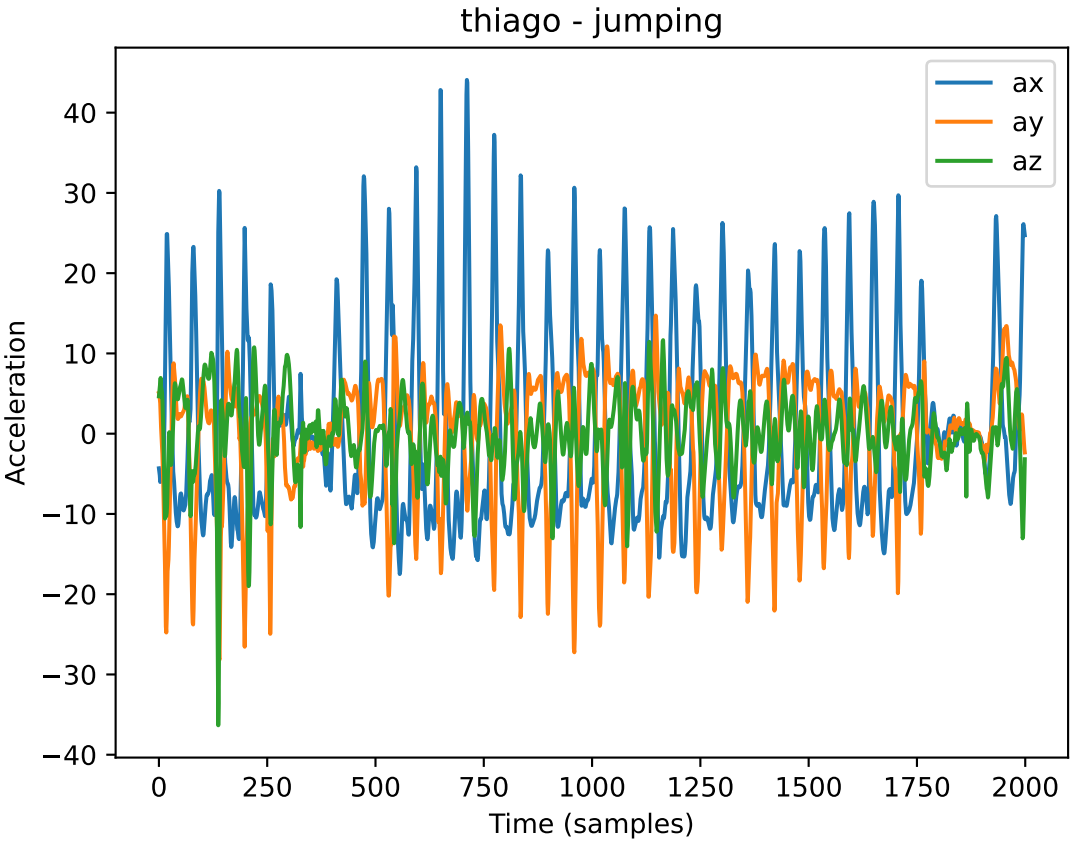

Walking vs Jumping Magnitude - thiago
walking
jumping
Acceleration Magnitude
Time (samples)
0
10
20
30
40
50
0
250
500
750
1000
1250
1500
1750
2000

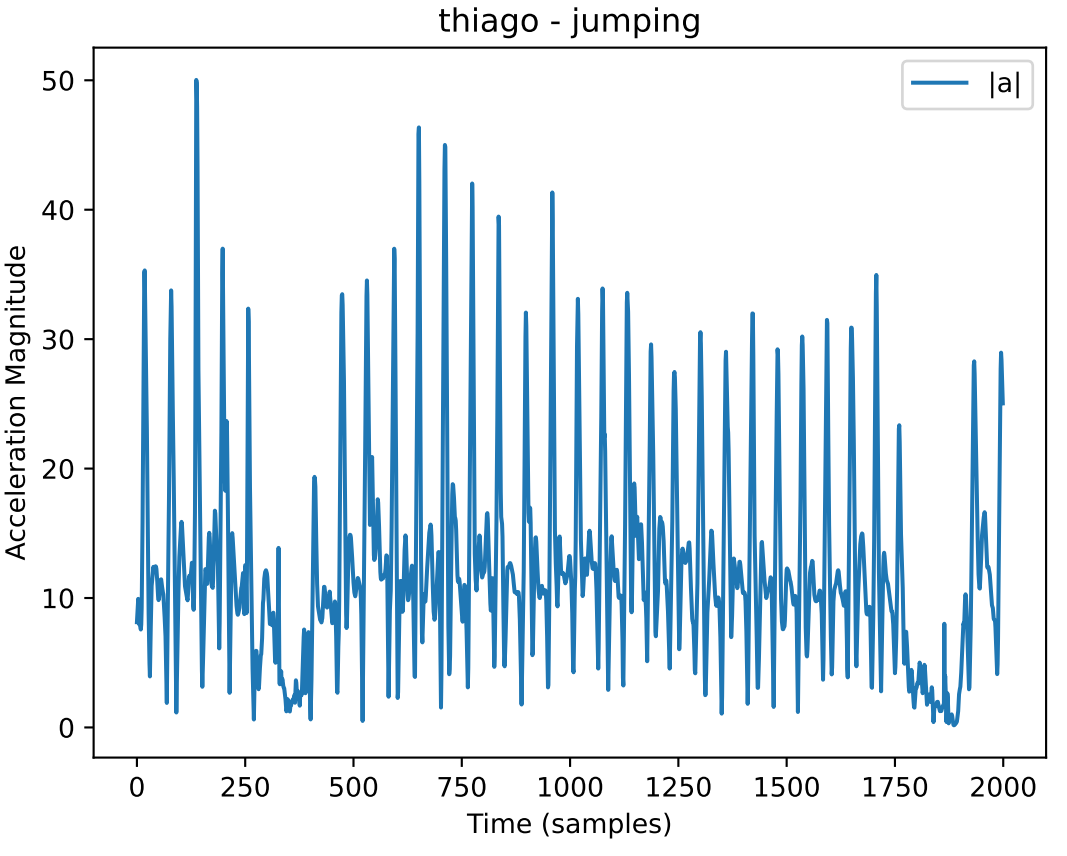

thiago - walking
ax
ay
az
Acceleration
Time (samples)
20
10
0
−10
−20
0
250
500
750
1000
1250
1500
1750
2000

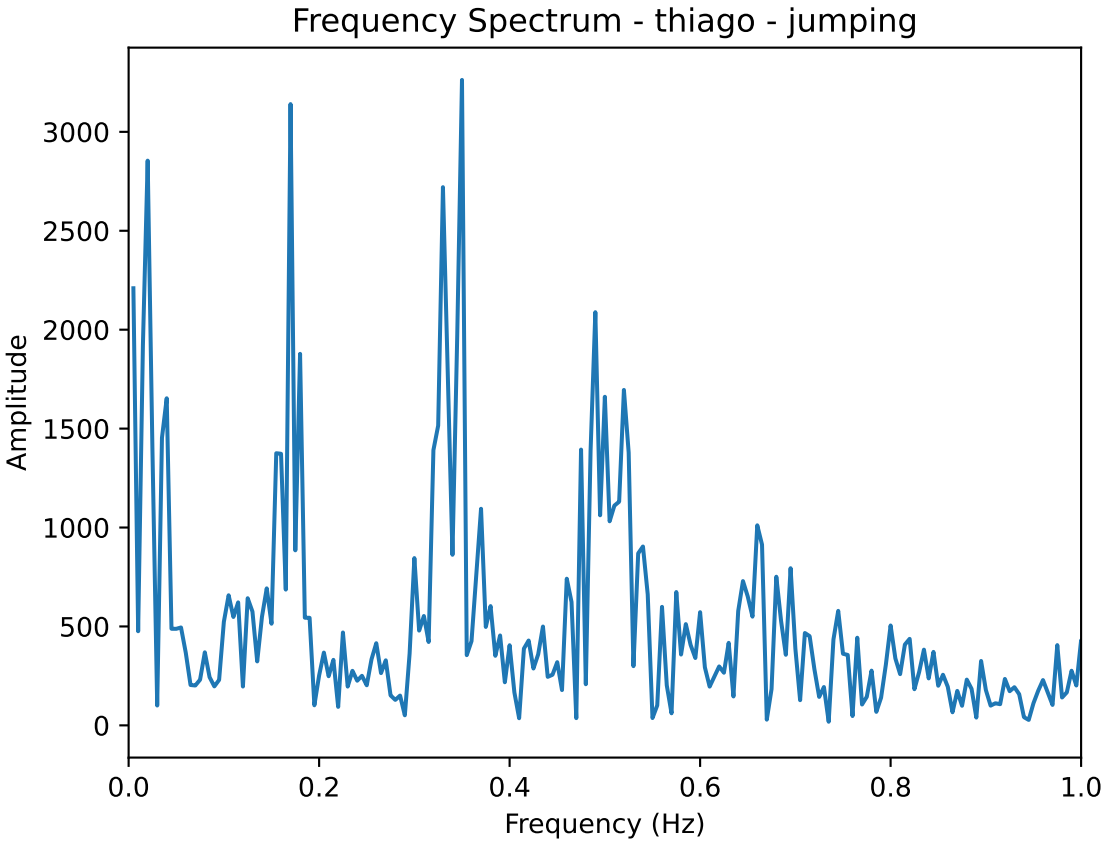

thiago - walking
|a|
Acceleration Magnitude
Time (samples)
0
5
10
15
20
25
0
250
500
750
1000
1250
1500
1750
2000

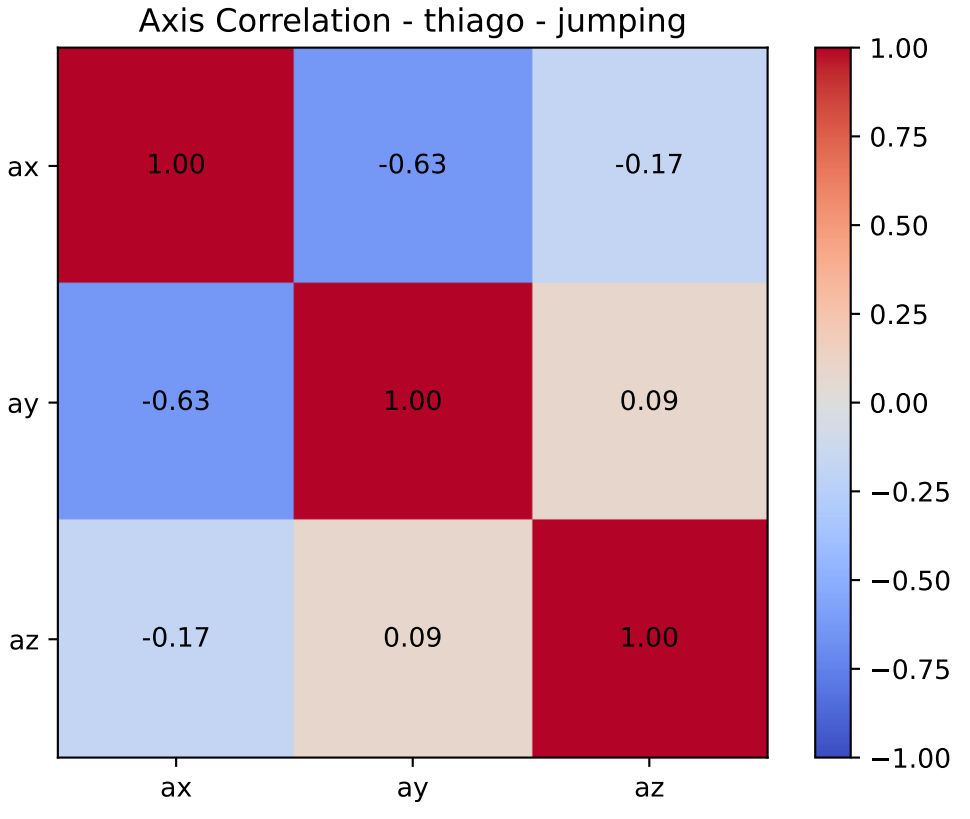

Frequency Spectrum - thiago - walking
Amplitude
1000
800
600
400
200
0
0.0
0.2
0.4
0.6
0.8
1.0
Frequency (Hz)

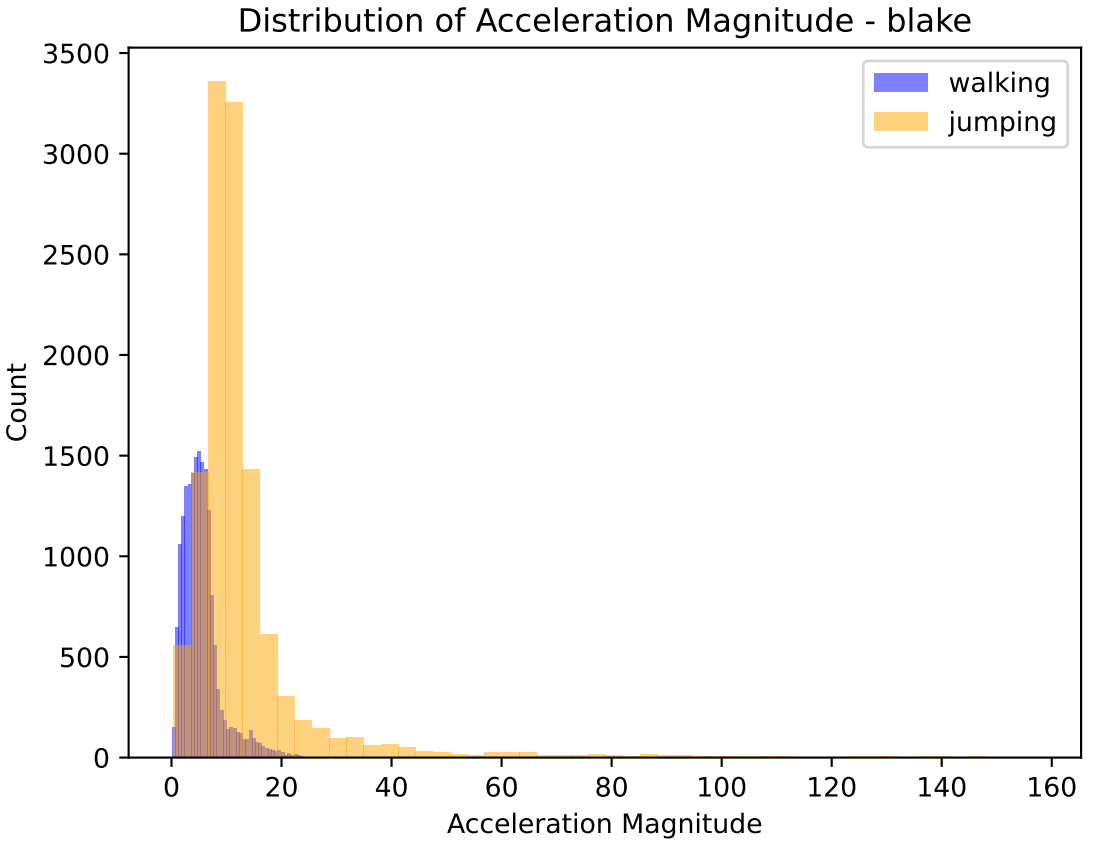

Axis Correlation - thiago - walking
ax
ay
az
1.00
0.19
0.45
0.19
1.00
0.15
0.45
0.15
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

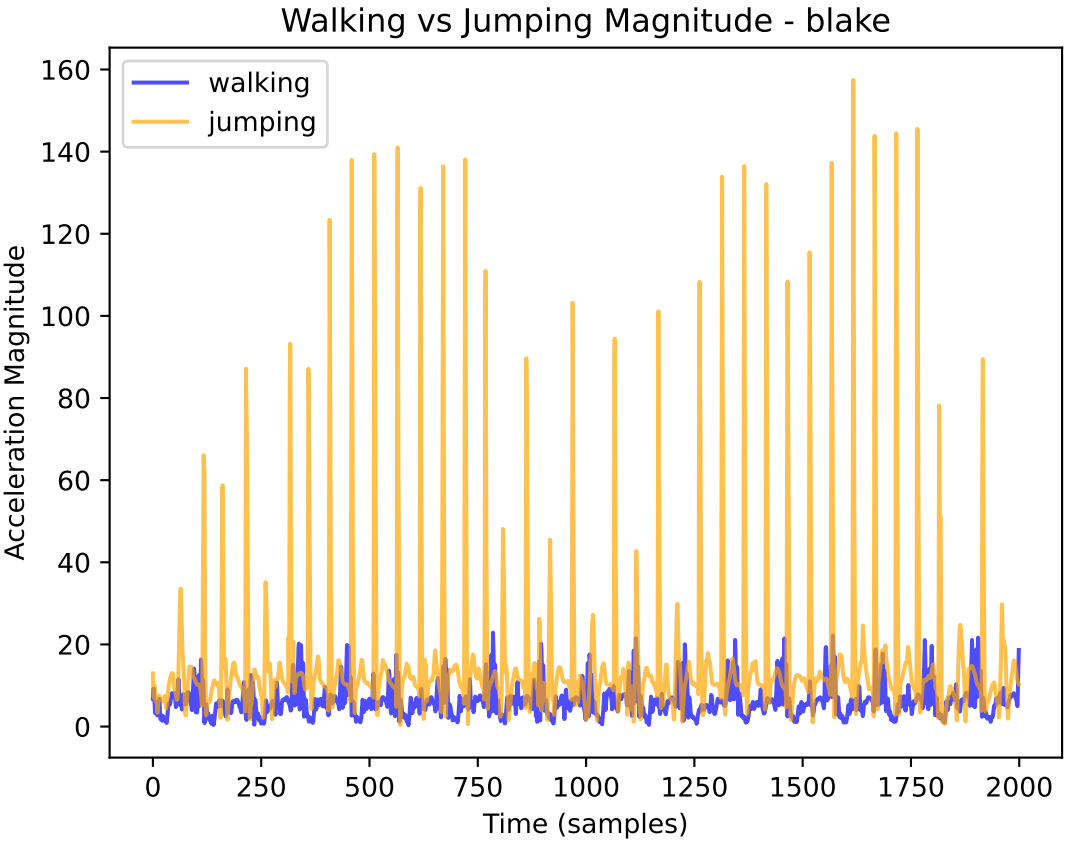

thiago - jumping
ax
ay
az
Acceleration
Time (samples)
40
30
20
10
0
−10
−20
−30
−40
0
250
500
750
1000
1250
1500
1750
2000

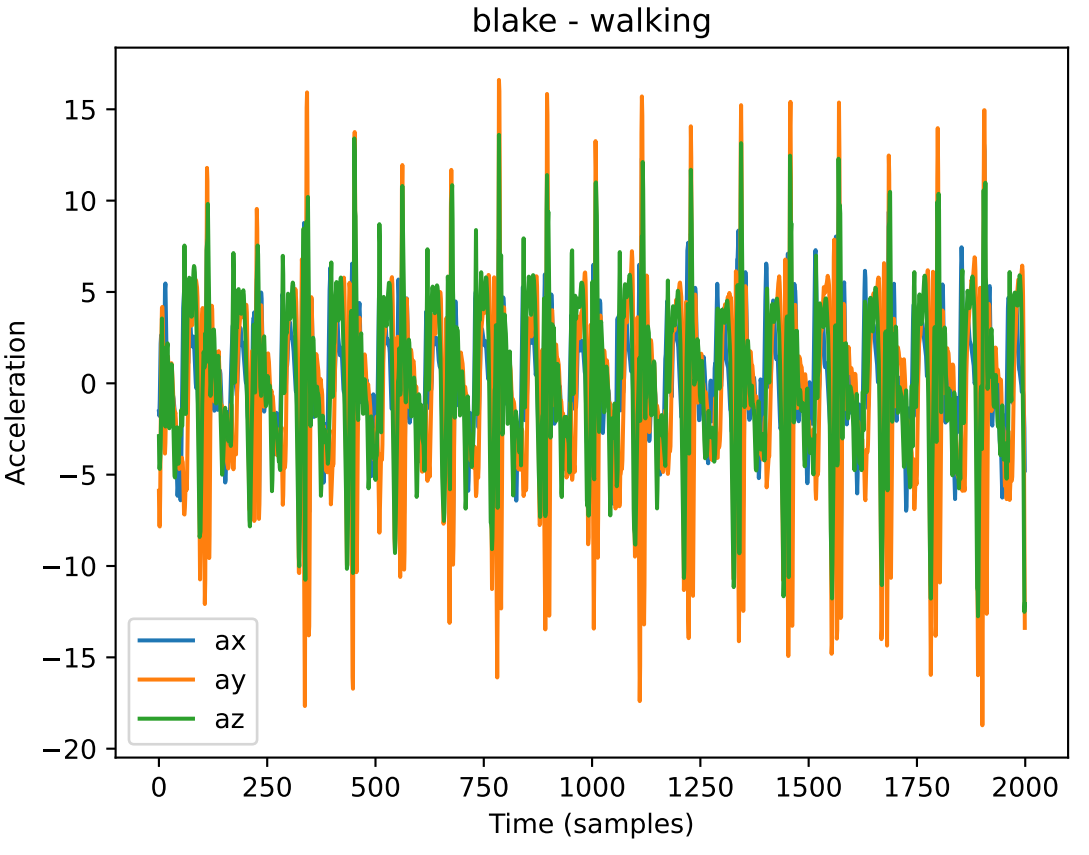

thiago - jumping
|a|
Acceleration Magnitude
Time (samples)
0
10
20
30
40
50
0
250
500
750
1000
1250
1500
1750
2000

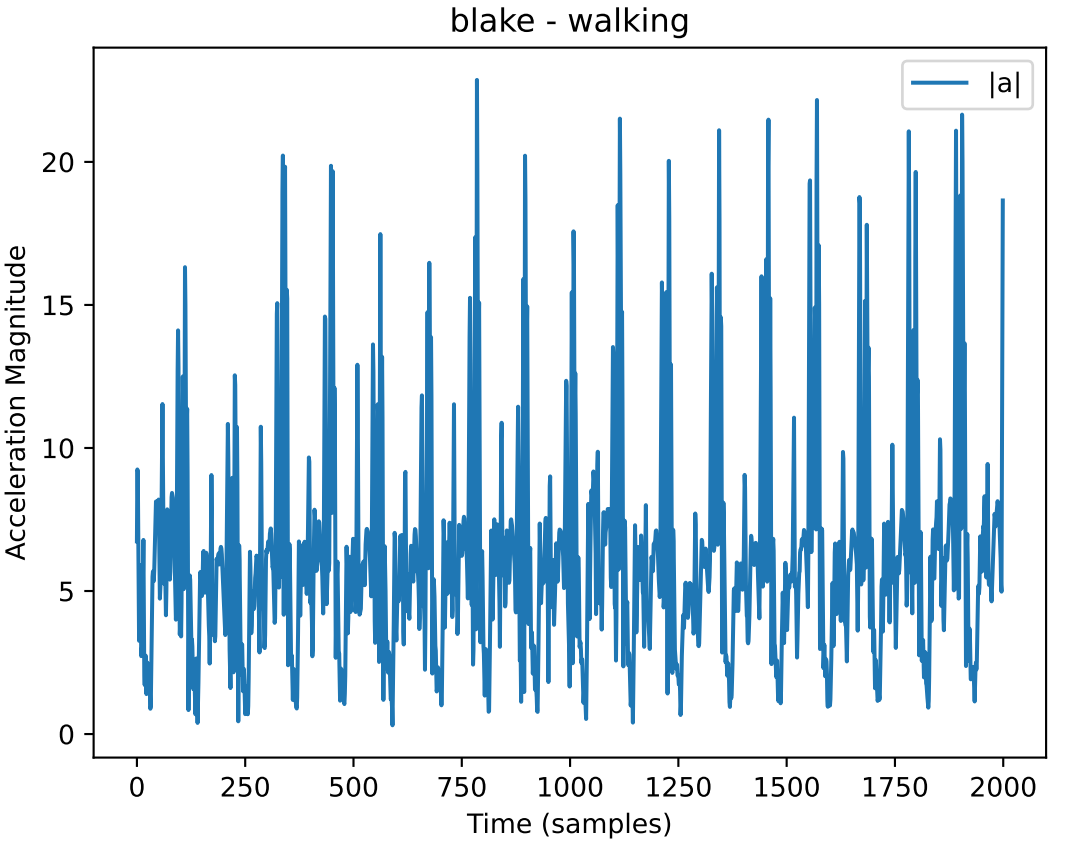

Frequency Spectrum - thiago - jumping
Amplitude
3000
2500
2000
1500
1000
500
0
0.0
0.2
0.4
0.6
0.8
1.0
Frequency (Hz)

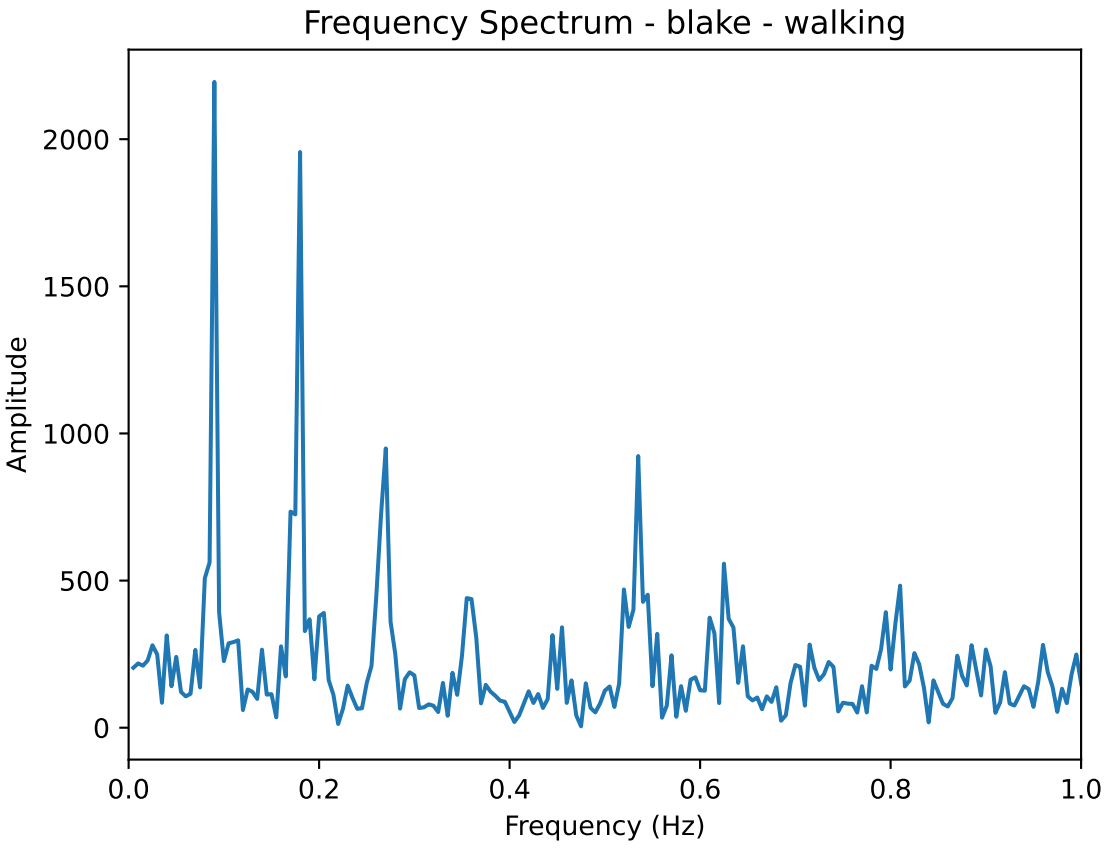

Axis Correlation - thiago - jumping
ax
ay
az
1.00
-0.63
-0.17
-0.63
1.00
0.09
-0.17
0.09
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

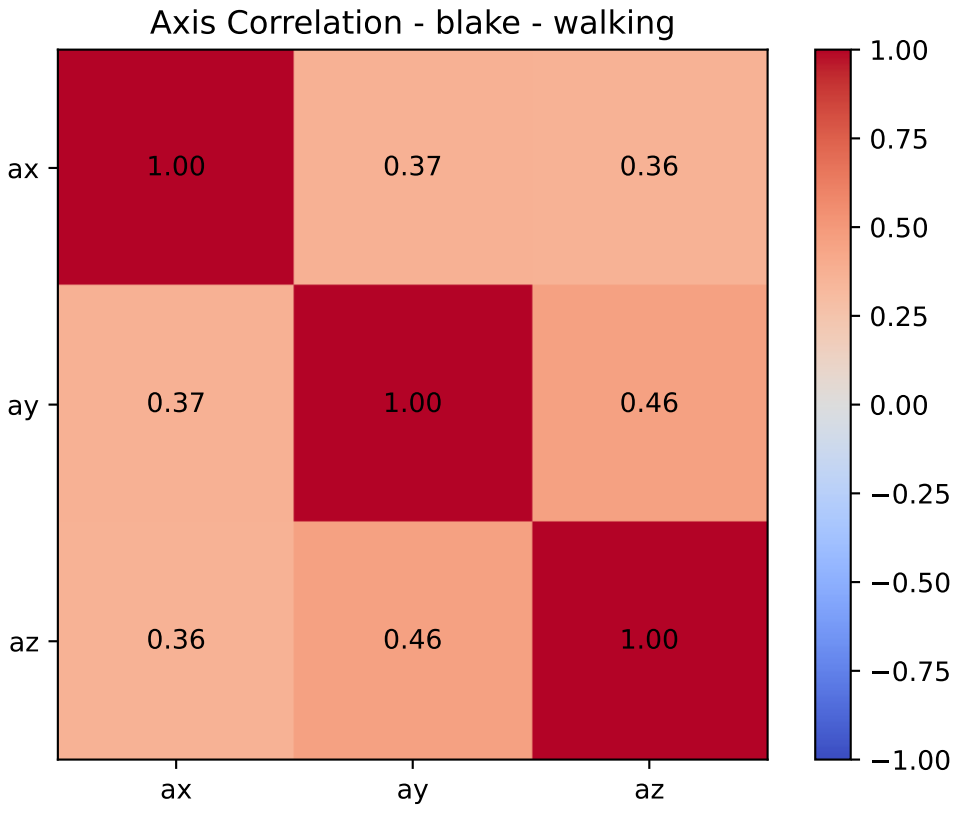

Distribution of Acceleration Magnitude - blake
walking
jumping
Count
Acceleration Magnitude
0
500
1000
1500
2000
2500
3000
3500
0
20
40
60
80
100
120
140
160

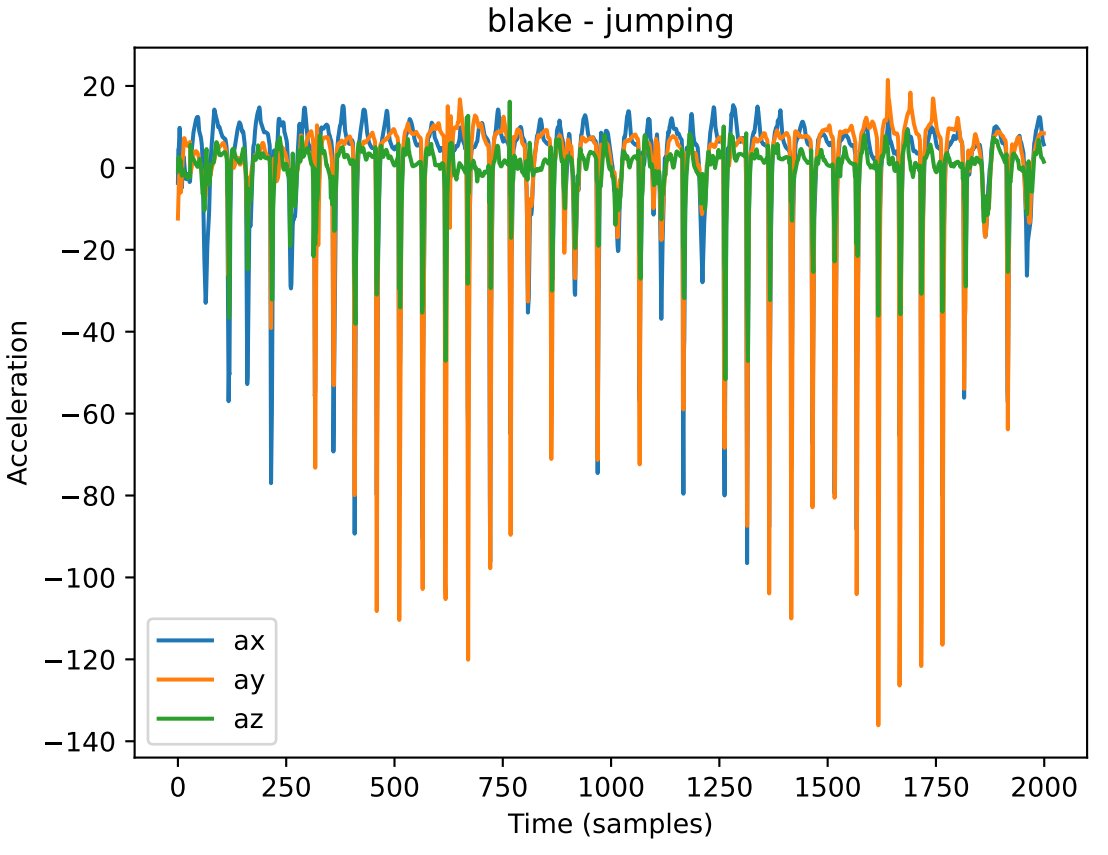

Walking vs Jumping Magnitude - blake
walking
jumping
Acceleration Magnitude
Time (samples)
0
20
40
60
80
100
120
140
160
0
250
500
750
1000
1250
1500
1750
2000

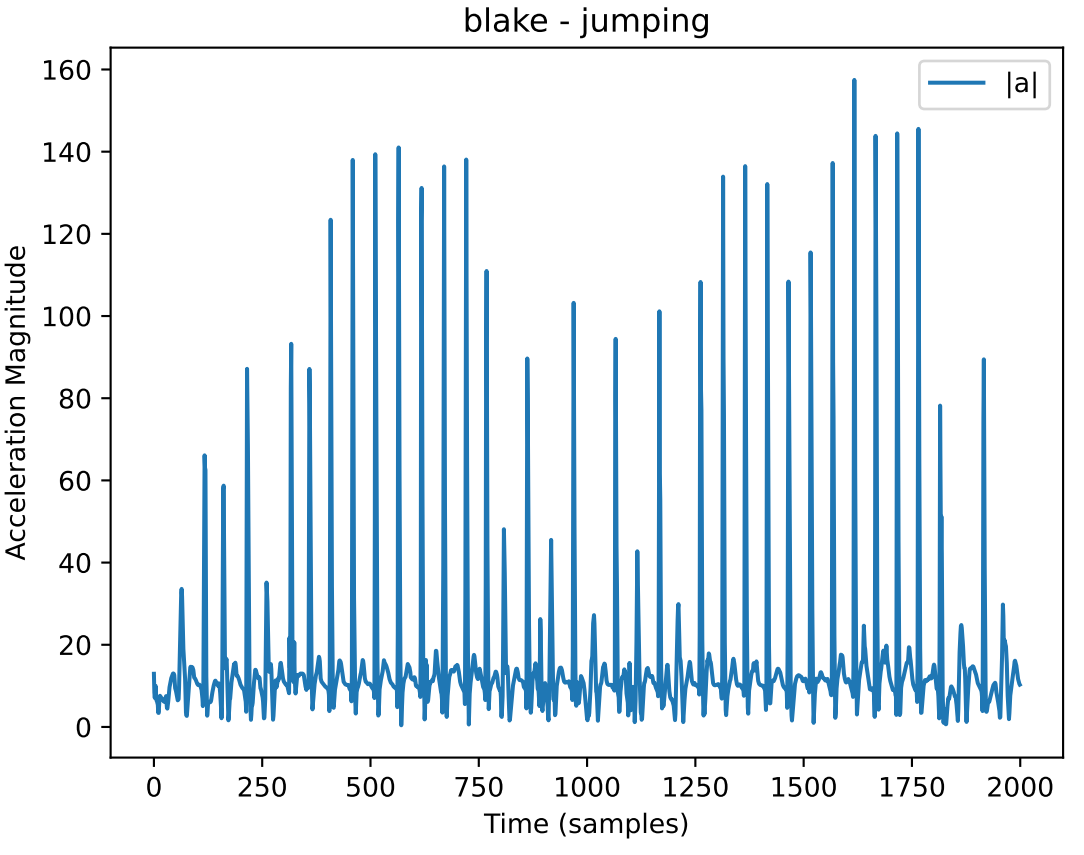

blake - walking
Acceleration
Time (samples)
ax
ay
az
15
10
5
0
−5
−10
−15
−20
0
250
500
750
1000
1250
1500
1750
2000

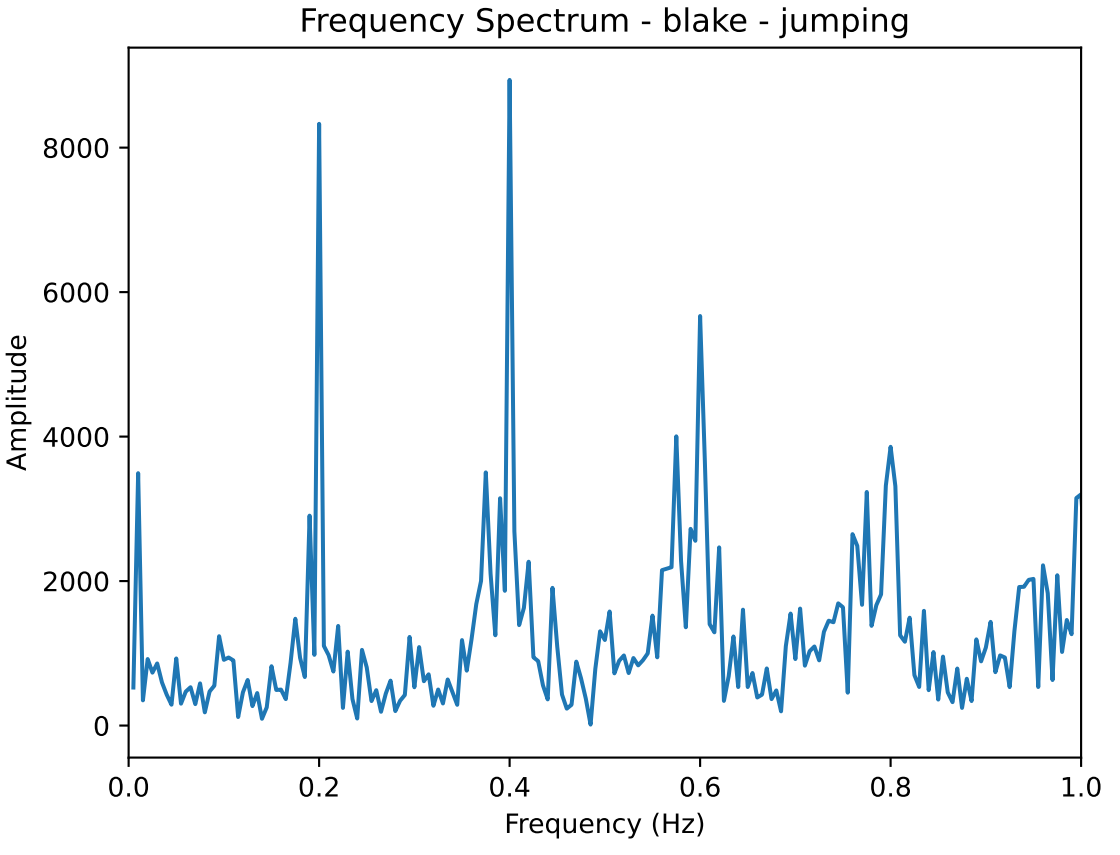

blake - walking
|a|
Acceleration Magnitude
Time (samples)
0
5
10
15
20
0
250
500
750
1000
1250
1500
1750
2000

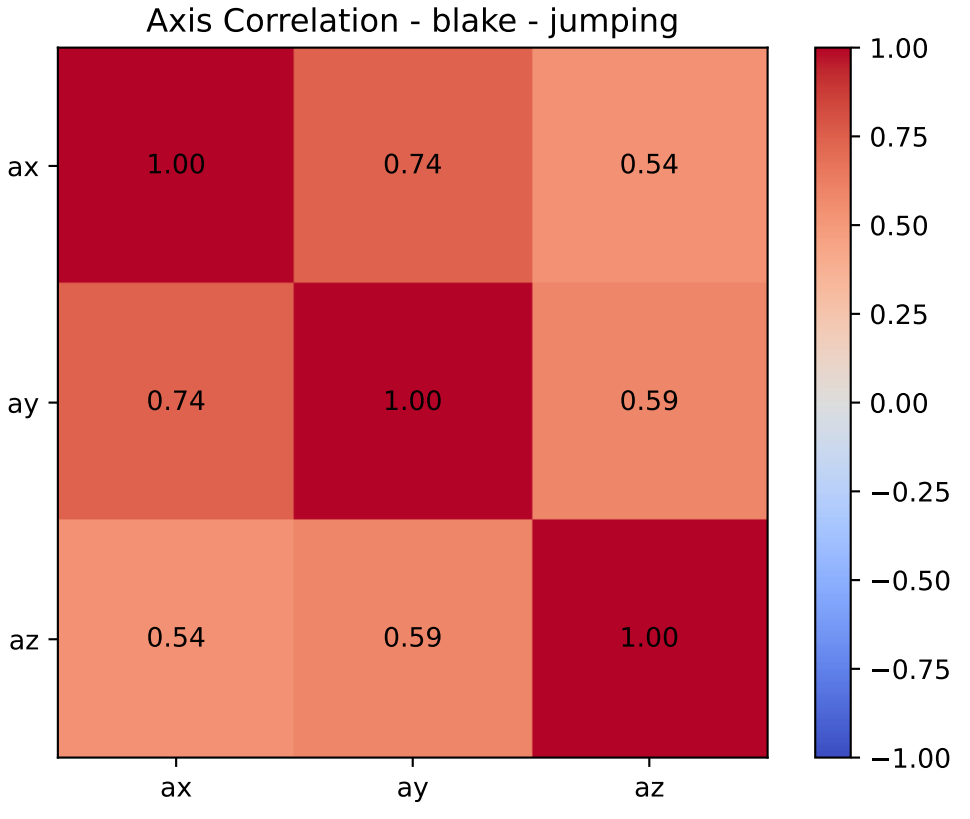

Frequency Spectrum - blake - walking
Amplitude
2000
1500
1000
500
0
0.0
0.2
0.4
0.6
0.8
1.0
Frequency (Hz)

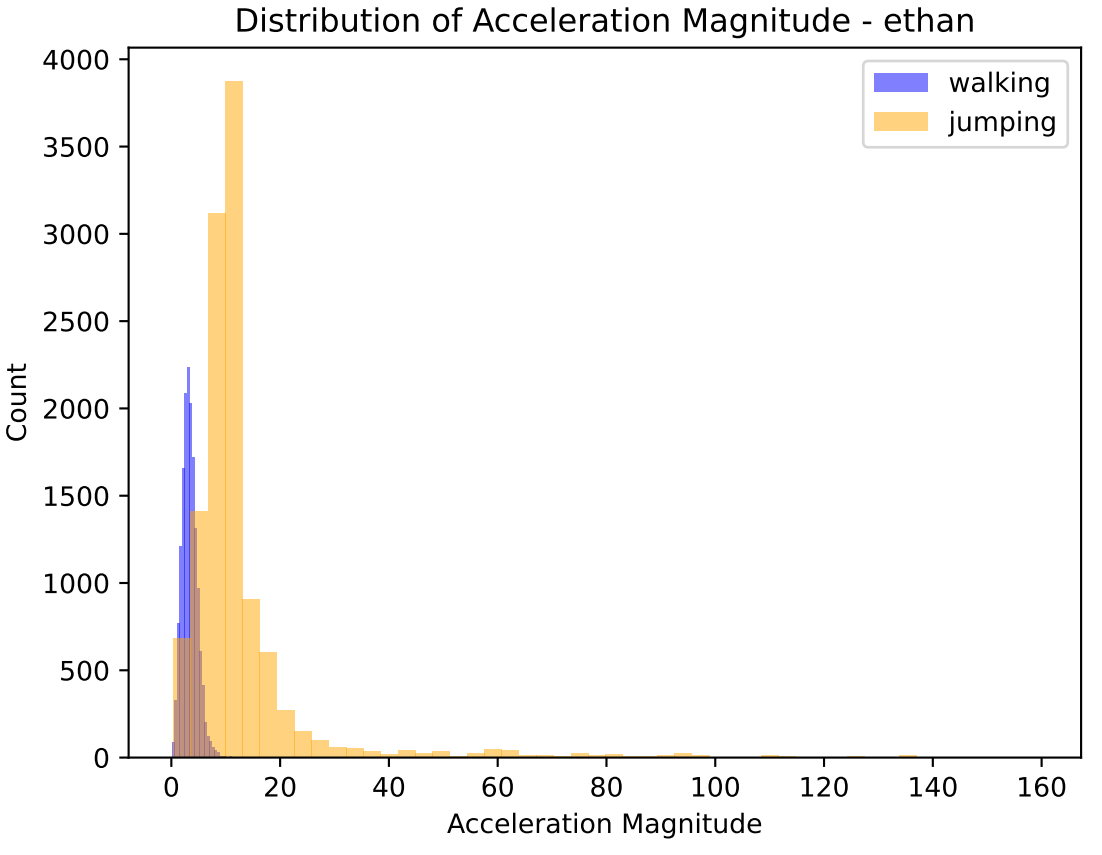

Axis Correlation - blake - walking
ax
ay
az
1.00
0.37
0.36
0.37
1.00
0.46
0.36
0.46
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

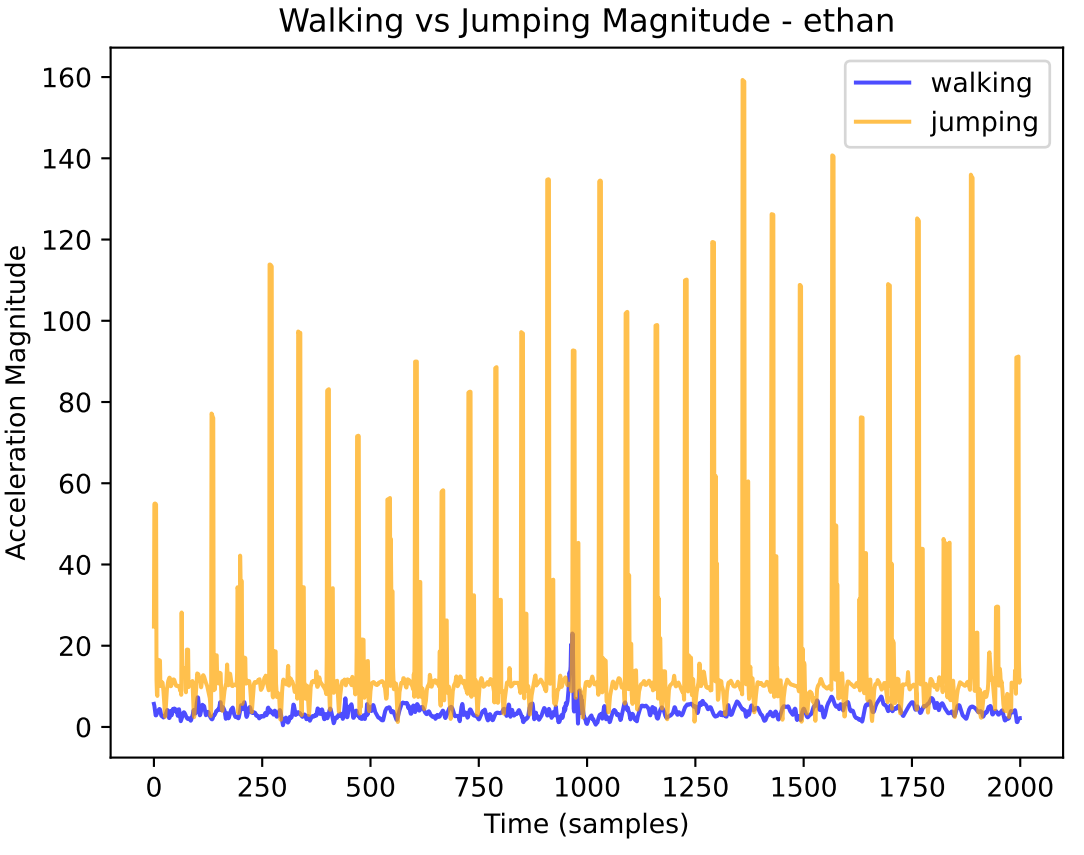

blake - jumping
Acceleration
Time (samples)
20
0
−20
−40
−60
−80
−100
−120
−140
0
250
500
750
1000
1250
1500
1750
2000
ax
ay
az

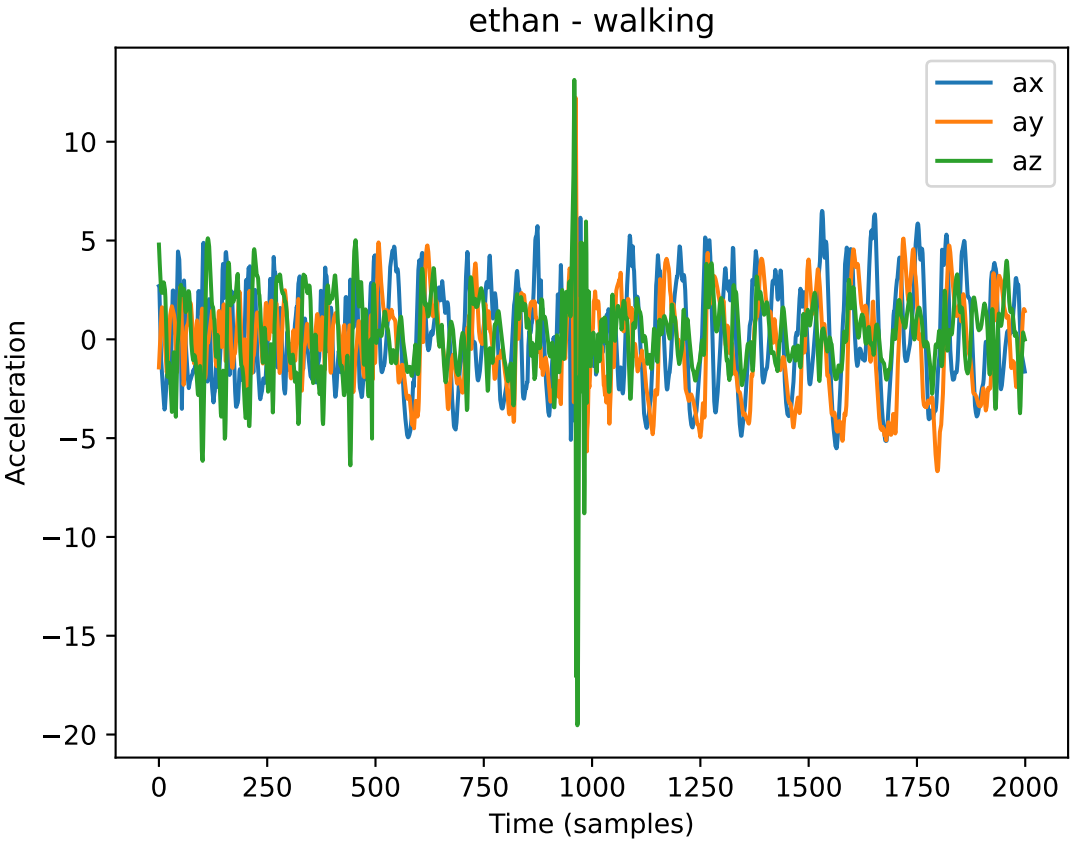

blake - jumping
|a|
Acceleration Magnitude
Time (samples)
0
20
40
60
80
100
120
140
160
0
250
500
750
1000
1250
1500
1750
2000

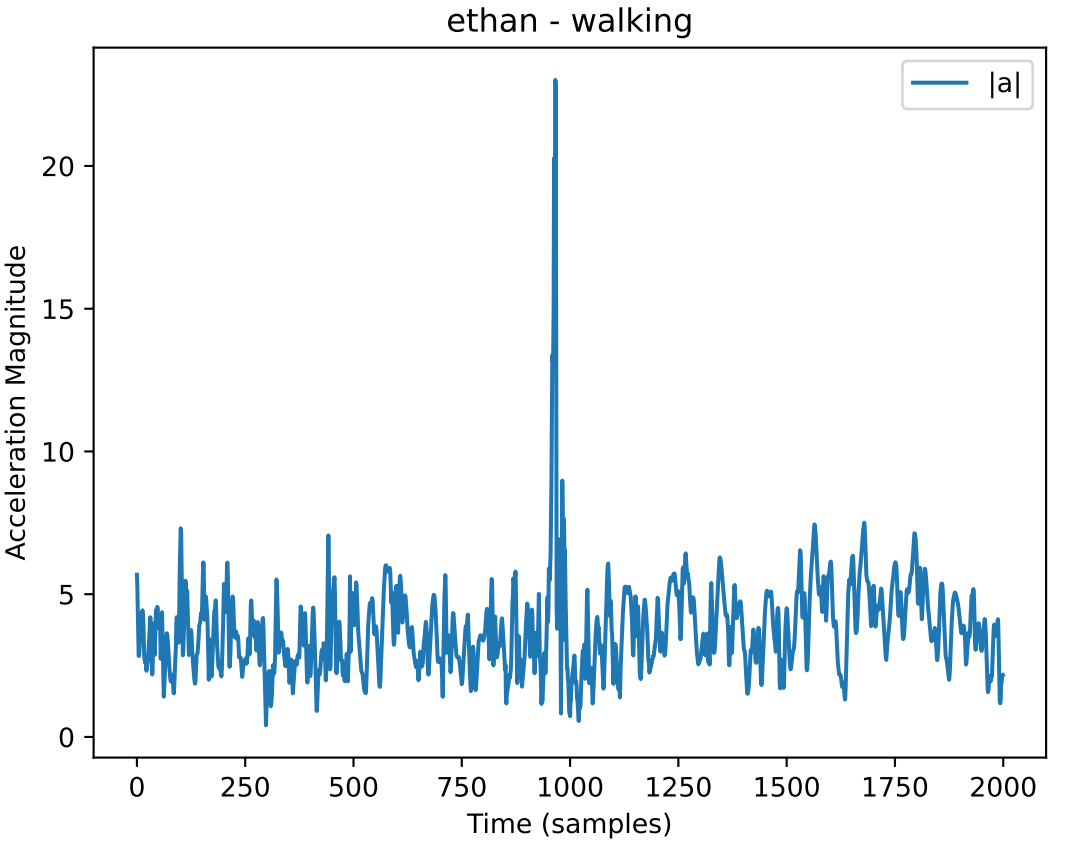

Frequency Spectrum - blake - jumping
Amplitude
8000
6000
4000
2000
0
0.0
0.2
0.4
0.6
0.8
1.0
Frequency (Hz)

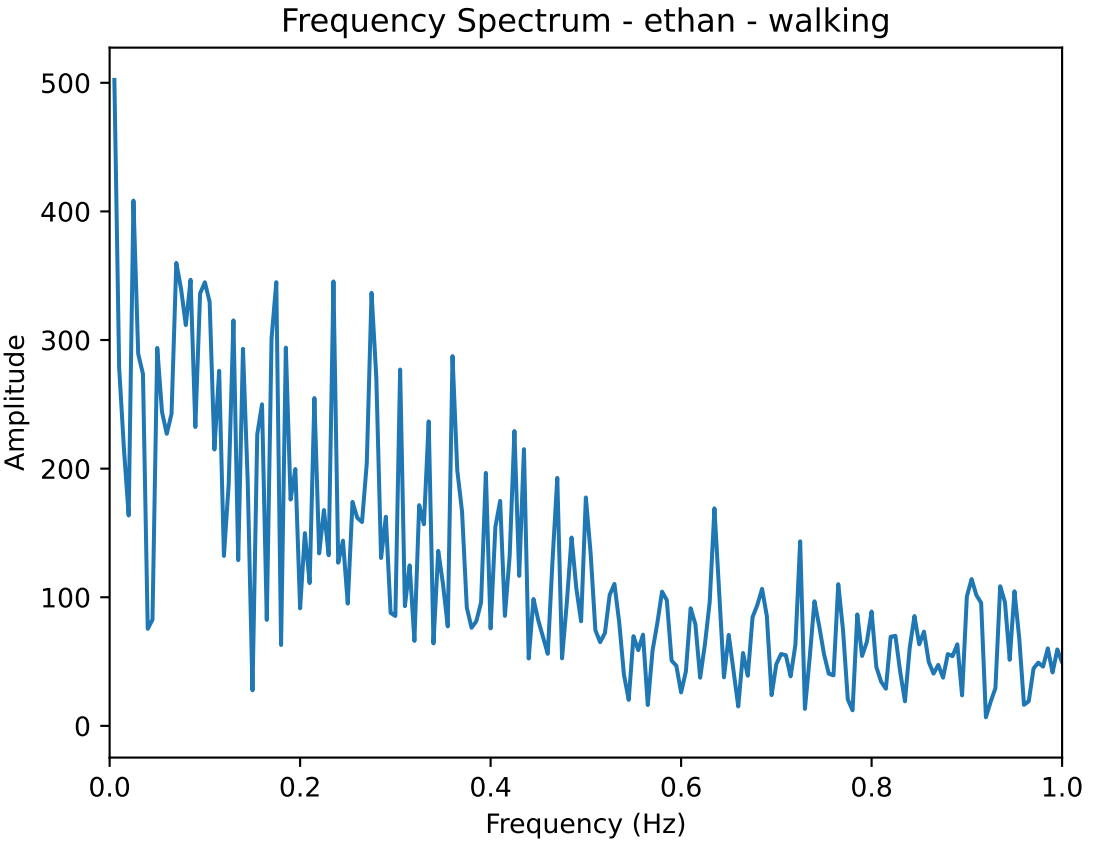

Axis Correlation - blake - jumping
ax
ay
az
1.00
0.74
0.54
0.74
1.00
0.59
0.54
0.59
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

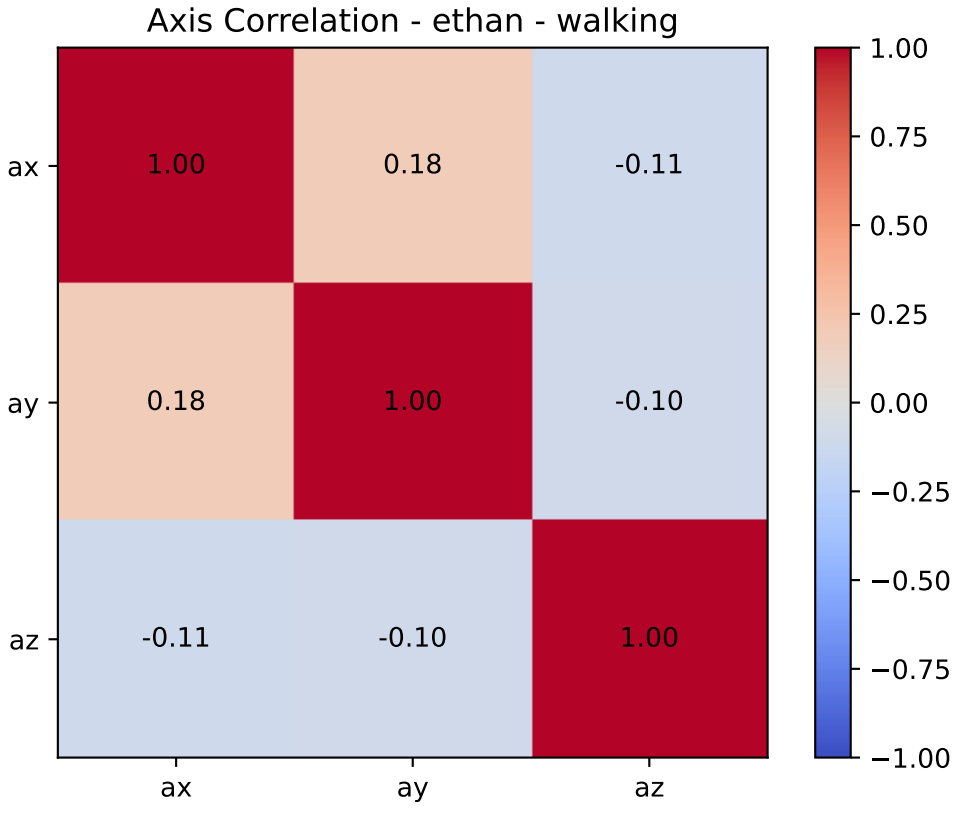

Distribution of Acceleration Magnitude - ethan
walking
jumping
Count
Acceleration Magnitude
0
500
1000
1500
2000
2500
3000
3500
4000
0
20
40
60
80
100
120
140
160

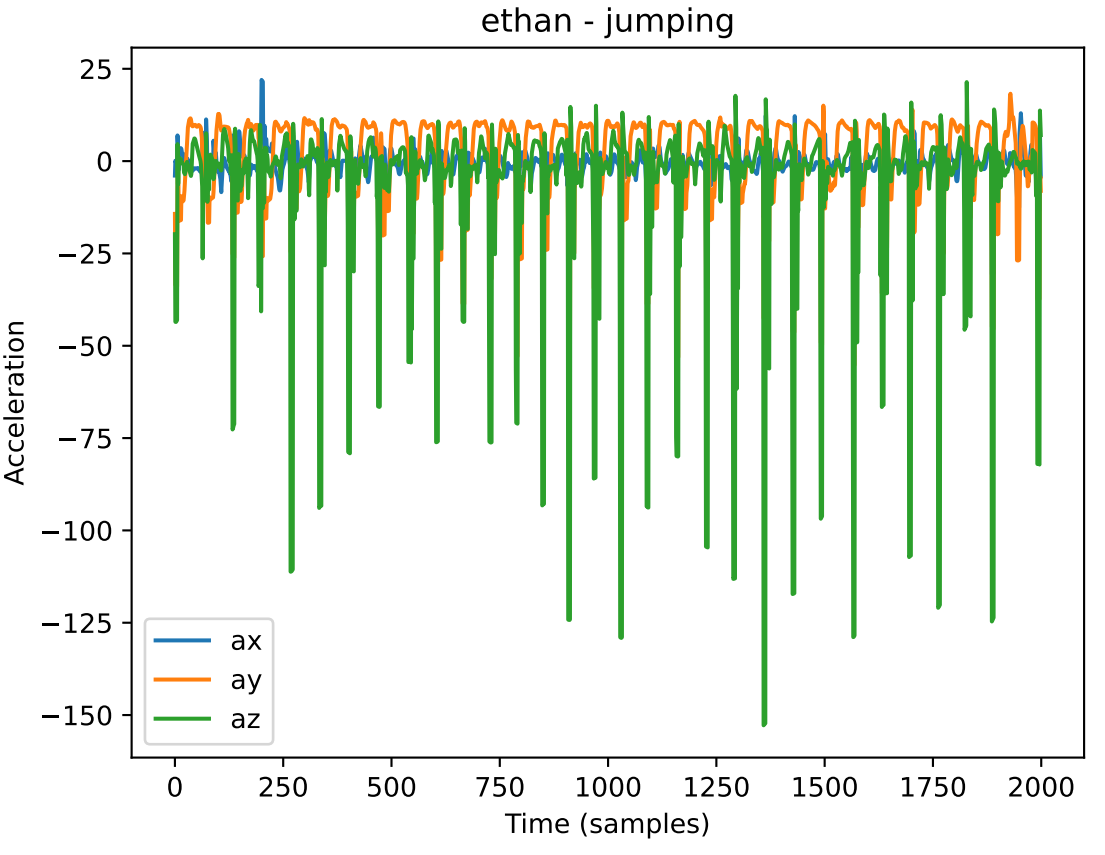

Walking vs Jumping Magnitude - ethan
walking
jumping
Acceleration Magnitude
Time (samples)
0
20
40
60
80
100
120
140
160
0
250
500
750
1000
1250
1500
1750
2000

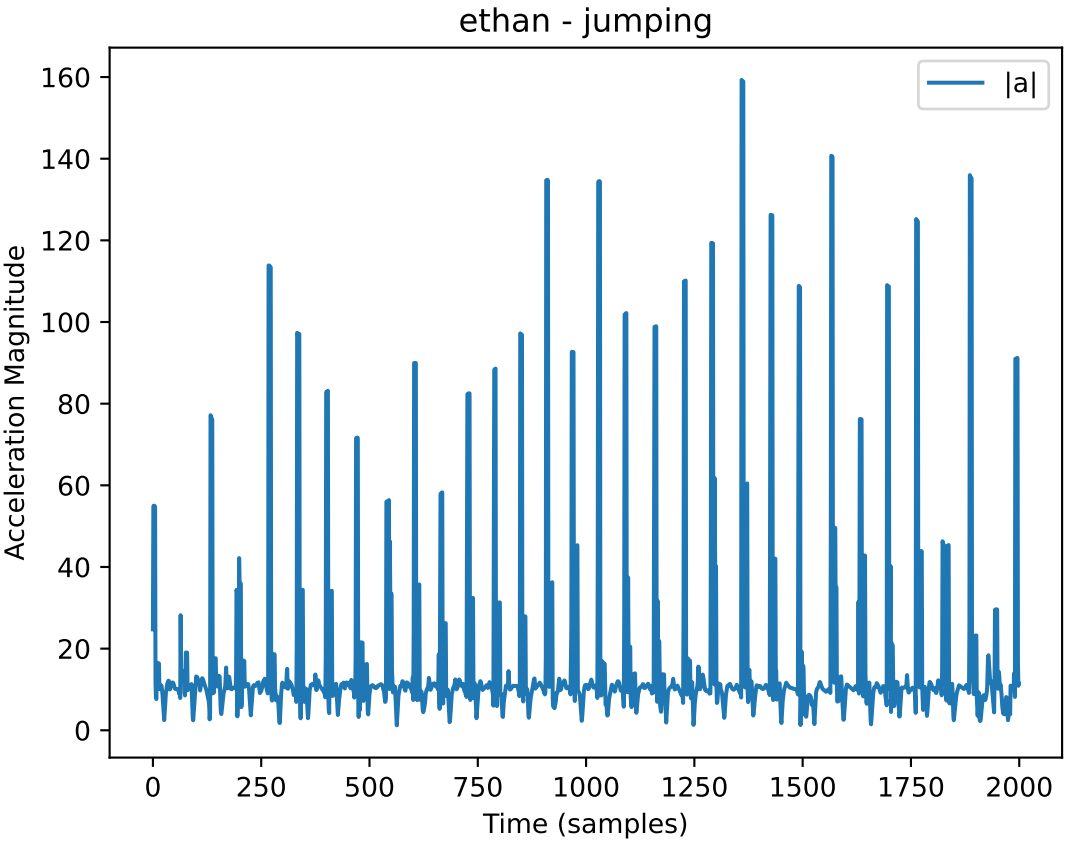

ethan - walking
ax
ay
az
10
5
0
−5
−10
−15
−20
Acceleration
0
250
500
750
1000
1250
1500
1750
2000
Time (samples)

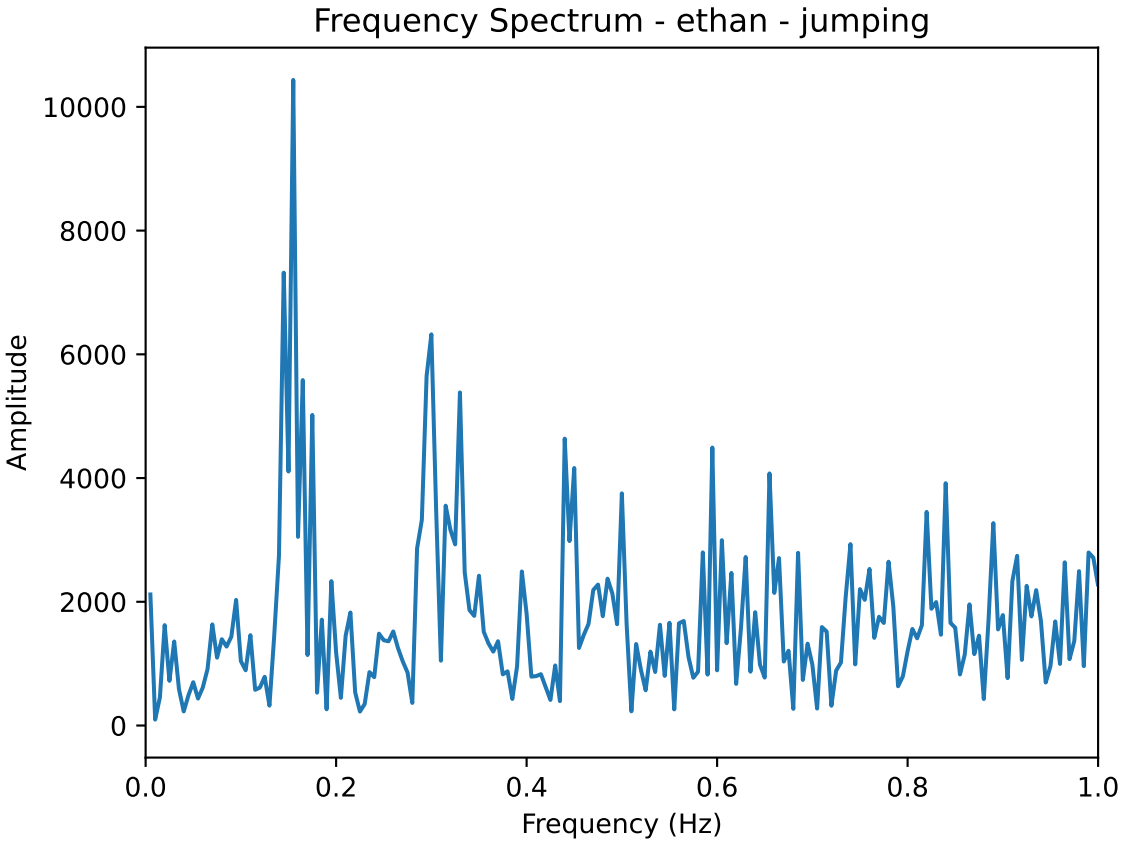

ethan - walking
|a|
Acceleration Magnitude
0
5
10
15
20
0
250
500
750
1000
1250
1500
1750
2000
Time (samples)

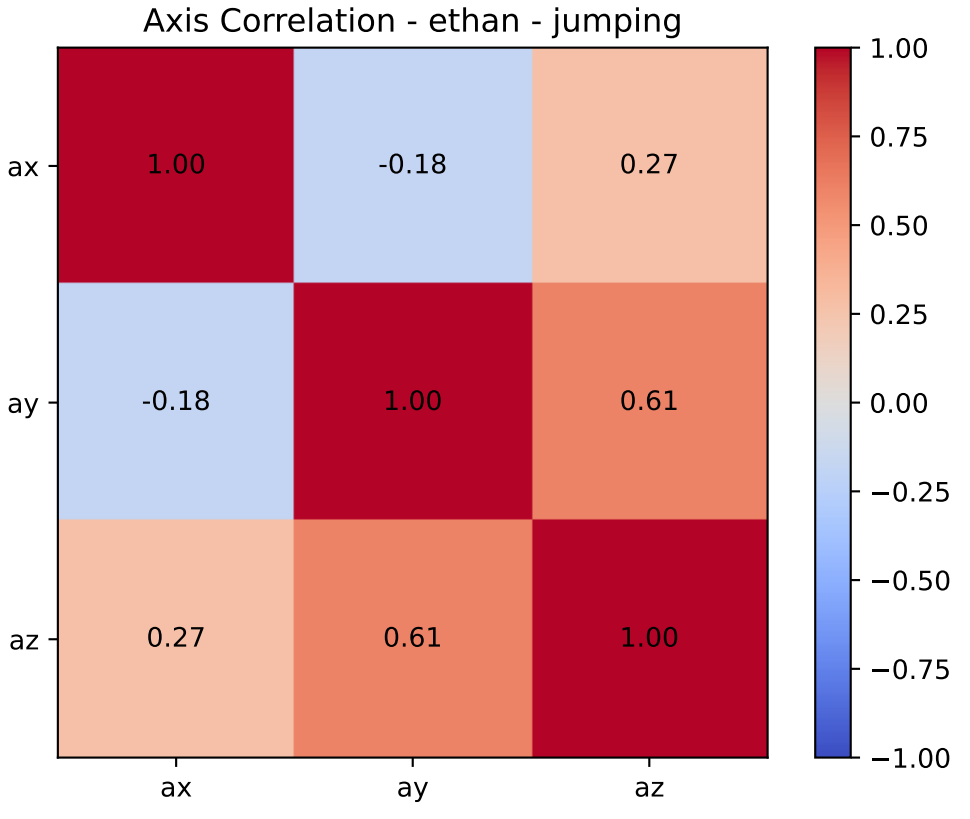

Frequency Spectrum - ethan - walking
Amplitude
Frequency (Hz)
500
400
300
200
100
0
0.0
0.2
0.4
0.6
0.8
1.0


Axis Correlation - ethan - walking
ax
ay
az
1.00
0.18
-0.11
0.18
1.00
-0.10
-0.11
-0.10
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


ethan - jumping
Acceleration
Time (samples)
ax
ay
az
25
0
−25
−50
−75
−100
−125
−150
0
250
500
750
1000
1250
1500
1750
2000


ethan - jumping
|a|
Acceleration Magnitude
Time (samples)
0
20
40
60
80
100
120
140
160
0
250
500
750
1000
1250
1500
1750
2000


Frequency Spectrum - ethan - jumping
Amplitude
10000
8000
6000
4000
2000
0
0.0
0.2
0.4
0.6
0.8
1.0
Frequency (Hz)


Axis Correlation - ethan - jumping
ax
ay
az
1.00
-0.18
0.27
-0.18
1.00
0.61
0.27
0.61
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00